pfi

Python for Fixed Income

COURSE 3 · MARKET DATA AND TIME SERIES

Slopes and Butterflies

WEEK 2 · DAY 4

Created by **Jeff Lenamon**

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The shape of a curve: slopes and flies

A slope

One yield less another, in bp. 2s10s is the 10 Yr less the 2 Yr.

A butterfly (a fly)

The belly against the two wings. 2s5s10s is 2 x 5 Yr less 2 Yr less 10 Yr.

1. Three slopes and their signs
2. Two flies, and the other sign
3. slope_bp and butterfly_bp, tested
4. Since 2015, with z-scores
5. DV01 weights from bondmath
6. Live: FRED's T10Y2Y

Every number today describes the shape of the curve on a date and how it changed. Nothing says what comes next.

Slopes: named short then long

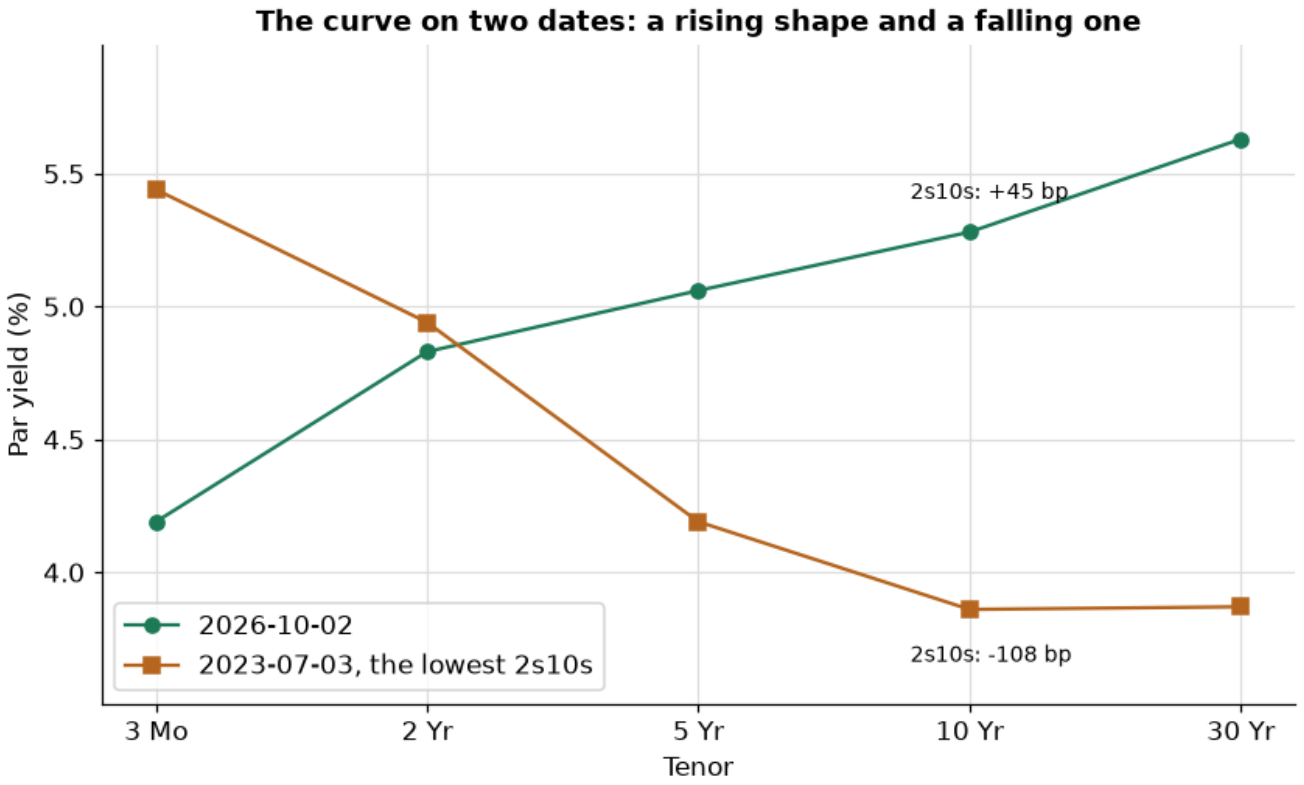
Name	Formula (yields in percent)	Positive means	2026-10-02
2s10s	(10 Yr - 2 Yr) x 100	10 Yr above 2 Yr	45 bp
5s30s	(30 Yr - 5 Yr) x 100	30 Yr above 5 Yr	57 bp
3m10y	(10 Yr - 3 Mo) x 100	10 Yr above 3 Mo	109 bp

2s10s went from 41 bp on 30 Sep 2026 to 45 bp on 2 Oct 2026. Both yields fell; the 2 Yr fell more.

Lowest in the file: -108 bp on 3 Jul 2023. The curve fell from the 2-year point to the 10-year point.

Desks call a negative slope inverted. Here the word describes the shape on a date and nothing more.

A slope is the direction of a line



Two shapes

Green rises from the 2 Yr to the 10 Yr: 2s10s is 45 bp. Orange falls: -108 bp on 3 Jul 2023.

The sign is the direction of the line between the two tenors.

Butterflies: twice the belly less the wings

```
fly_bp = (2 x belly - short - long) x 100 = 2 x (belly - average of the wings) x 100
```

1 bp

2s5s10s: the 5 Yr half a bp above the line from the 2 Yr to the 10 Yr

-13 bp

5s10s30s: the 10 Yr below the average of the 5 Yr and 30 Yr

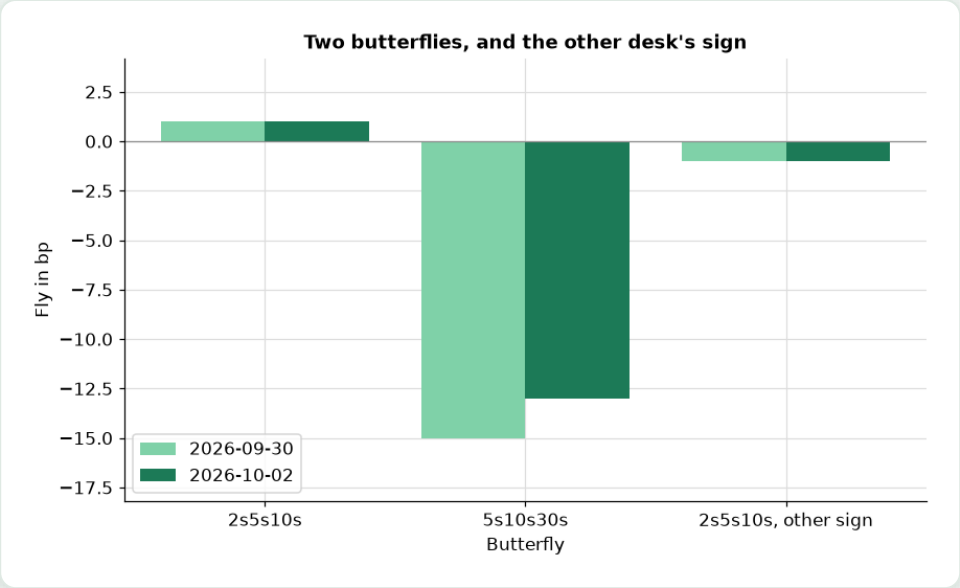
-1 bp

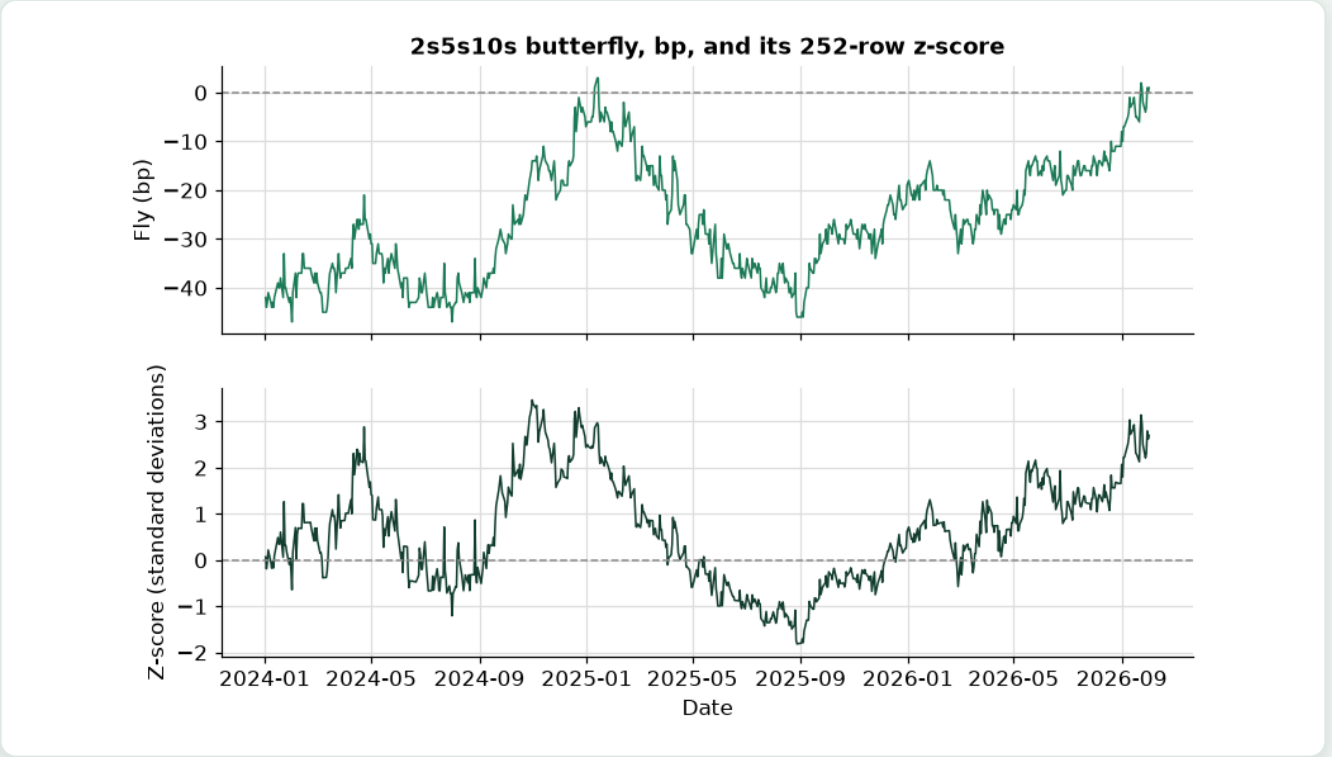
2s5s10s on a desk that quotes wings less belly: same curve

Some desks quote the opposite sign. Ask which one before comparing a fly with anyone else's.

Slopes and flies are separate facts: 5s30s was 57 bp while 5s10s30s was -13 bp.

Butterflies, drawn





Left: bars above zero mean the belly yields more than the average of the wings. Right: the fly and its z-score since 2024.

slope_bp and butterfly_bp

```
def slope_bp(curve, short, long):
    # long less short, percent to bp
    return (curve[long] - curve[short]) * 100
def butterfly_bp(curve, short, belly, long):
    # twice the belly, less each wing
    return (2*curve[belly] - curve[short] - curve[long]) * 100
```

Both functions also

check every tenor is a column (ValueError), check the dates (day 3), and name the result.

Test	What it pins down
`test_slope_sign`	positive and negative rows, both directions
`test_butterfly_hand_value`	(2 x 5.09 - 4.88 - 5.29) x 100 = 1 bp
`test_butterfly_matches_excel_reference`	190 rows of 2026, to 1e-9 bp

38 passed

Slopes and flies over time

544

of 2,940 market days with 2s10s below zero

537

of them in one run, 6 Jul 2022 to 26 Aug 2024

-0.51

z-score of 2s10s on 2 Oct 2026 (252 rows)

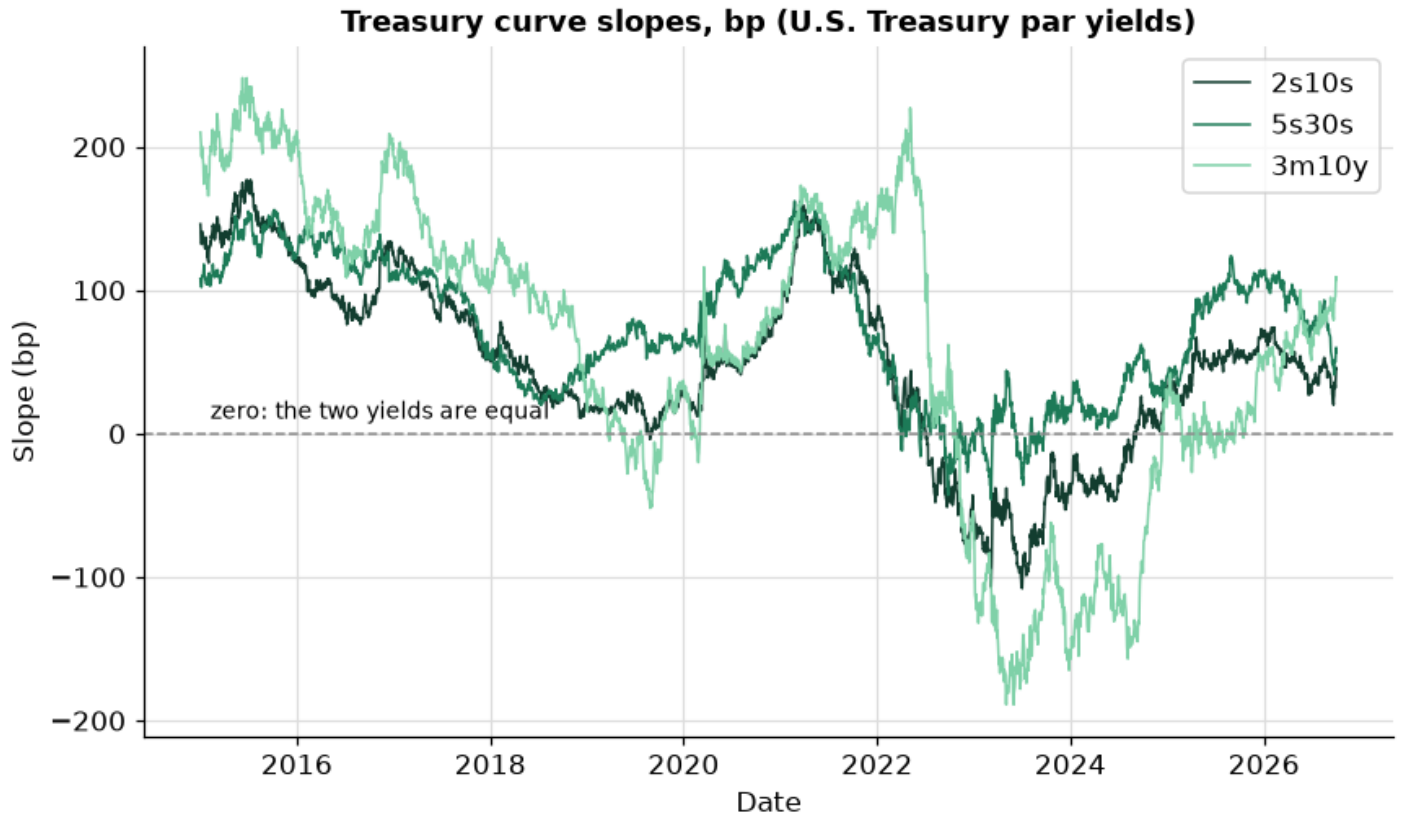
2.70

z-score of 2s5s10s on 2 Oct 2026 (252 rows)

The z-score of a slope is a z-score of a series: day 8's rolling_zscore equals Excel's STANDARDIZE column of the reference file on all 190 rows.

A z-score describes where today sits in its own last year. Never a signal, never a reason to expect a move.

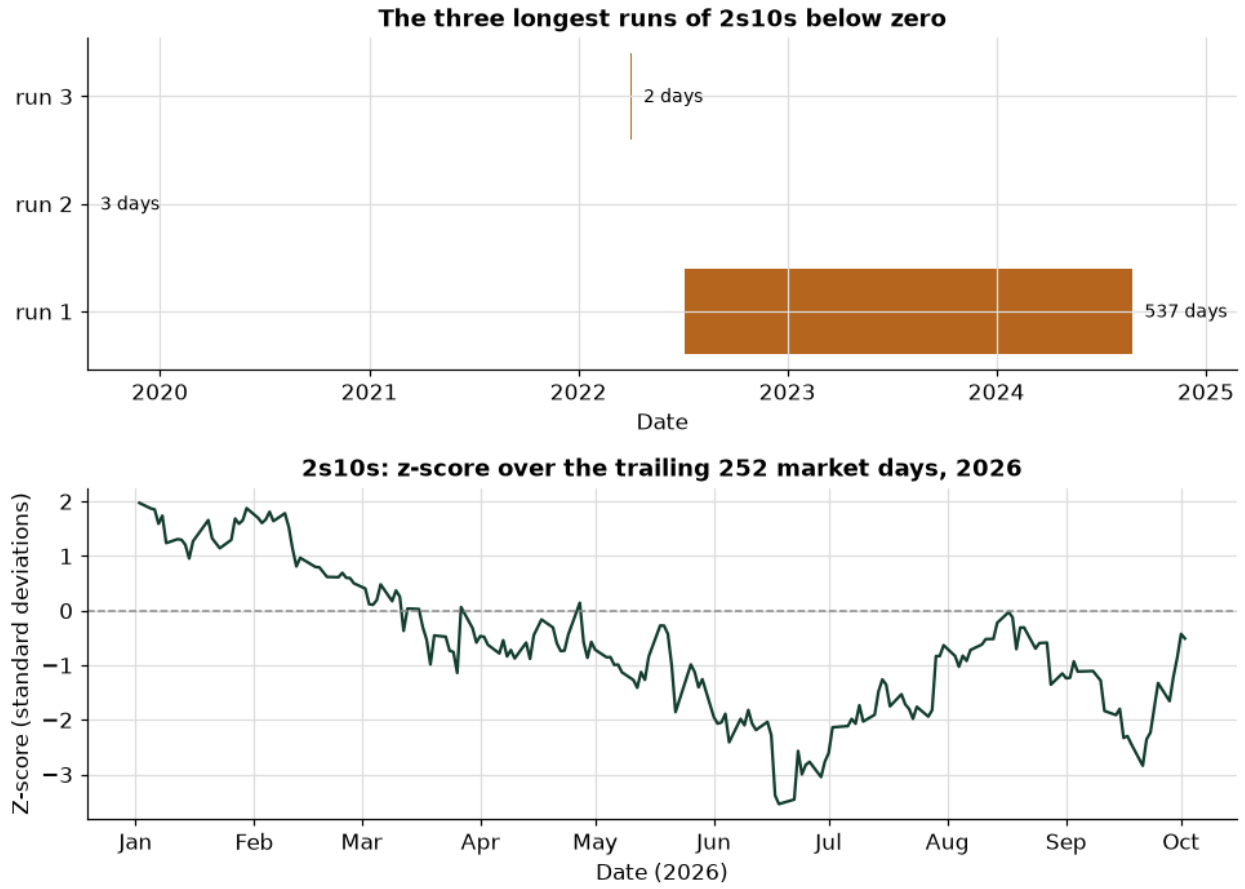
The three slopes since 2015, zero marked



Reading the lines

Above the dashed line the longer yield is higher. Below it the shorter one is. Each slope has its own history.

The longest negative runs, and 2026's z-score



Reading the two panels

Top: each bar is one unbroken run of negative days. One run is 537 market days. Bottom: where 2s10s sat in its own last year during 2026.

Both describe the history in the file.

DV01-neutral fly weights, from bondmath

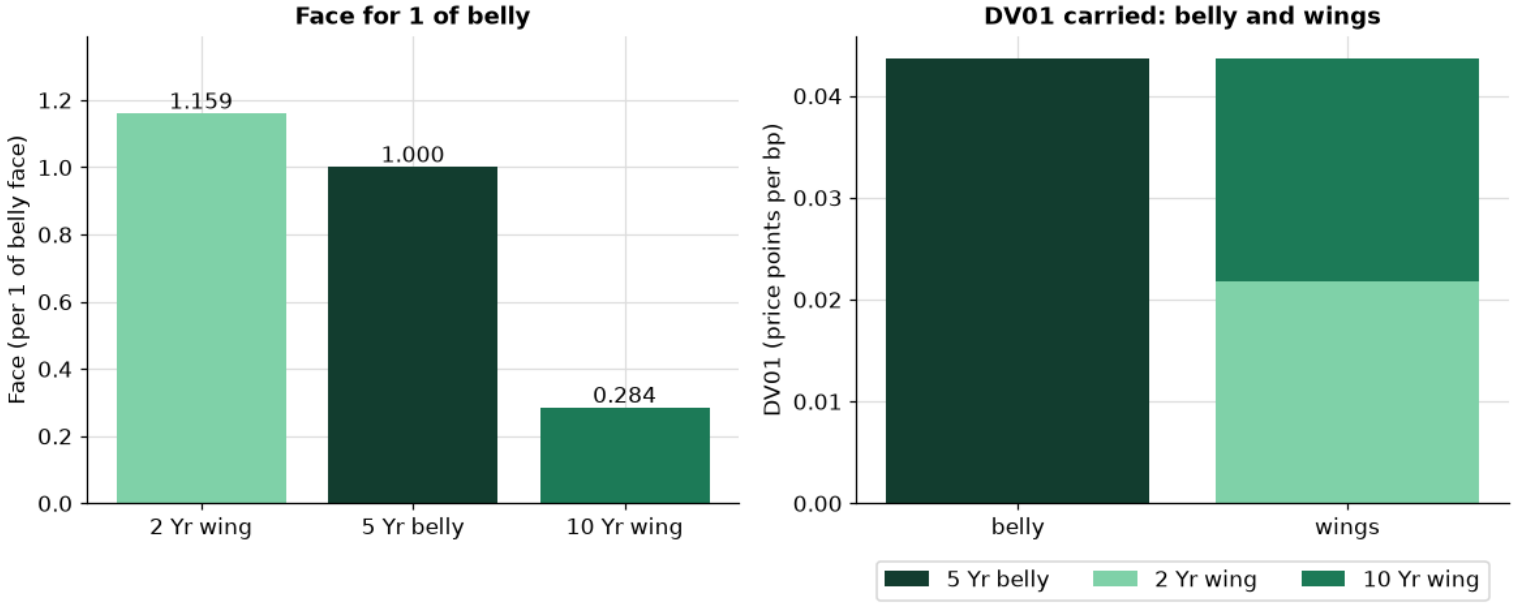
Par bond	Coupon = yield	Matures	DV01	Face per 1 of belly
2 Yr wing	4.83%	2028-10-02	0.018848	1.159049
5 Yr belly	5.06%	2031-10-02	0.043693	1.000000
10 Yr wing	5.28%	2036-10-02	0.076925	0.283995

w_short = 0.5 x DV01_belly / DV01_short, w_long = 0.5 x DV01_belly / DV01_long. DV01 per 100 of face per bp, settlement 2 Oct 2026.

Belly 0.043693 = 2 Yr wing 0.021846 + 10 Yr wing 0.021846: the DV01s net to zero

Arithmetic, not a position: the faces at which three DV01s net to zero. Never a size or a trade.

The faces, and the DV01s they carry



Reading the panels

Left: the short wing needs the most face. Right: the wings stacked reach the belly's height, so the DV01s net to zero.

Arithmetic, not a position.

Live: FRED's T10Y2Y against your 2s10s

T10Y2Y is the 10-year Treasury constant maturity yield less the 2-year, in percent, from the same Treasury yields as the snapshot.

```
live_t10y2y = md.fred_series("T10Y2Y", fred_key, ...)
# yours in bp, FRED's in percent: divide yours by 100
assert live_gap <= 1e-9
```

A live cell, two ways

No key (the saved notebook):

```
Live cell skipped: no FRED key was found.
```

With your key:

row and dot counts, dates, the check

Three checks: every snapshot date has a value, FRED has no extra date, the largest gap is within 1e-9 percent.

This product uses the FRED API but is not endorsed or certified by the Federal Reserve Bank of St. Louis.

Excel side by side

Shape	Excel, row 2, filled down	2026-10-02
2s10s	`=(M2-I2)*100`	45 bp
5s30s	`=(O2-K2)*100`	57 bp
3m10y	`=(M2-E2)*100`	109 bp
2s5s10s	`=(2*K2-I2-M2)*100`	1 bp
5s10s30s	`=(2*M2-K2-O2)*100`	-13 bp

DV01, Course 2's way

```
=MDURATION(...)*(PRICE(...)+acc)
  /10000
```

5-year: MDURATION 4.369256, DV01 0.043693. Weights =0.5*C4/B4 and =0.5*C4/D4.

Columns in the Treasury file: E 3 Mo, I 2 Yr, K 5 Yr, M 10 Yr, O 30 Yr. Check the header row before filling down.

Tested in Excel 16.0 build 20527, 2026-11-02: every row of the file equal to Python's.

Save the day: a data file in Git

1. Commit

marketdata.py and the tests

2. Write, commit

slopes_2026.csv, two decimals

3. Mistake

flip one sign, then git diff

4. Restore

git restore: back as committed

```
-2026-09-30,41.00,55.00,109.00
+2026-09-30,-41.00,55.00,109.00
```

Git diffs a CSV line by line: one line out, one line in. Read it before you commit a rerun.

The AI check

The docstring is the prompt. The code that comes back runs, and has one data bug.

1

Docstring

Units, conventions, and what it must never do: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: fly_bp writes (2 * belly - (short - long)) * 100, adding the long wing back: 1059 bp instead of 1 bp on the hand row.

Code that runs is not the same as code that is right.

What you can do now

- Slopes named short then long, positive when the longer yield is higher
- Flies as twice the belly less the wings; some desks use the other sign
- slope_bp and butterfly_bp, tested by hand and against Excel
- A slope's history and its 252-row z-score, described
- DV01-neutral weights from bondmath: arithmetic, never a position

Week 2, Day 5 Case study: three years of curve and spreads in one summary table